Lecture 10

Monday Feb. 24 2026

CPU Scheduling

- OS handles the scheduling to decide which process gets access to the CPU

Scheduling Criteria

- CPU utilization
- Throughput
- Turnaround time
- Waiting time
- Response time

Priority Scheduling

- A priority number (int) is associated with each process
- CPU is allocated to the process with the highest priority
  - Smallest integer = highest priority
  - preemptive
  - non-preemptive
- SJS is a priority scheduling where priority is the predicted next CPU burst time

Problem:
- **Starvation** - low priority processes may never execute

- Solution:
  - **Aging** - as time progresses increase the priority of the process

Example

- Scheduling example done in class for preemptive and non-preemptive priority scheduling
- Slide 33

Round Robin

- Each process gets a small unit of CPU time (time quantum), usually 10-100 milliseconds. After this time elapses.
- The process is preempted and added to the end of the ready queue

- If there are *n* processes in the ready queue and the time quantum is *q*, then each process gets 1/*n* of the CPU time in chunks of at most *q* time units at once. No process waits more than (*n*-1)*q* time units.
- Performance
- *q* large =>FIFO
- *q* small => *q* must be large with respect to context switch, otherwise overhead is too high
- Preemptive FCFS based on a timeout interval, the quantum q
- The running process is interrupted by the clock and put last in a FIFO "ready" queue
- Then the first "ready" process is run instead.
- Can be done with FCFS or SJS (Priority queue)

Round Robin Example

- Slide 40